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"Finite-ensemble observability of the mean-field convergence rate in consensus-based optimization"

The manuscript studies whether the theoretically expected mean-field J^{-1} scaling for consensus-based optimization can be observed from finite-ensemble numerical experiments, with particular emphasis on the roles of dimension, concentration of the Boltzmann weights, and the diffusion parameters. The question is interesting and potentially relevant to the numerical interpretation of the existing CBO theory. The author has also carried out an unusually extensive computational study and has made a serious effort to identify and disclose weaknesses in earlier versions of the experimental design.

However, I do not believe that the manuscript is ready for publication in its present form. In particular, there is an error in the assumption audit that directly affects one of the main conceptual conclusions, and the calibration procedure used for the principal corrected experiment changes the algorithm from realization to realization. Addressing these issues appears to require more than a revision of the exposition and, in my view, likely requires rerunning some of the central experiments. I therefore recommend rejection of the present manuscript, while I would be open to considering a substantially revised version as a new submission.

The principal issues are as follows.

1. The classification of the Ackley function in Table 1 appears to be incorrect. In particular, the manuscript states that Ackley satisfies condition (6),

$$|E(v) - E(w)| \leq C_1(\|v\|_2 + \|w\|_2)\|v - w\|_2.$$

   For the standard Ackley function, take $v = te_1$ and $w = 0$. As $t \to 0$, the first exponential term gives

$$E(te_1) - E(0) = \frac{4}{\sqrt{d}}|t| + O(t^2),$$

   whereas the right-hand side of (6) is $C_1 t^2$. Hence (6) cannot hold in a neighborhood of the origin.

   This is not a minor classification issue. Section 7.12 emphasizes that three configurations satisfying the parameter condition nevertheless fail the operational rate-consistency criterion, and all three of these examples use Ackley. Earlier in the paper, however, the author explains that the existence result accompanying the cited mean-field theorem requires (6). Thus the manuscript has not established that these Ackley experiments are actually examples for which all hypotheses needed for the cited theorem are satisfied. The resulting statement that the "theorem-admissible" region and the finite-run observability region are different sets therefore needs to be reconsidered. If a different well-posedness result covers Ackley, it should be stated explicitly and its assumptions should be checked. Otherwise this central conclusion should be demonstrated using objectives for which the complete theoretical assumptions are verified.

2. The treatment of the isotropic and anisotropic parameter conditions should be made theoretically precise. The manuscript defines

$$r_{\text{iso}} = \frac{d\sigma^2}{2\lambda}, \qquad r_{\text{aniso}} = \frac{\sigma^2}{2\lambda},$$

and subsequently interprets $r = 1$ as the theoretical boundary in both cases. The precise theorem supporting each boundary, together with its assumptions, should be cited and audited separately. At present the text moves rather quickly between the two diffusion models while referring to the parameter condition from the classical theory. Since a major part of Sections 7.10–7.15 is organized around the distinction between the admissible and non-admissible regimes, the mathematical basis for this distinction needs to be completely unambiguous.

3. The "exact calibration" used for the main concentration experiment introduces a serious change in the estimand. In Section 7.13, β is solved separately on each realization's reference ensemble so that

$$\frac{\mathrm{ESS}_0(J_\infty)}{J_\infty} = \tau_0$$

holds exactly. Consequently, β is itself random and depends on the initial empirical sample. Different realizations therefore use different algorithmic parameters.

The manuscript acknowledges this point, but nevertheless bases the headline corrected dimension comparison on this experiment and compares the resulting slopes with fixed-parameter mean-field theory. This makes the interpretation difficult: the experiment is averaging over a distribution of algorithms rather than observing one fixed algorithm over repeated realizations.

A cleaner design would determine a single value

$$\beta = \beta(f, d, \rho_0, \tau_0)$$

from an independent and sufficiently large calibration sample, or by accurately estimating the population quantity

$$\tau = \frac{(\mathbb{E}w)^2}{\mathbb{E}[w^2]},$$

and then hold that same β fixed over every realization and every ensemble size in the corresponding cell. This would preserve the intended concentration matching without conditioning the algorithmic parameter on the particular realization. In my view, the principal dimension/concentration experiment should be repeated under such a fixed-β design before strong conclusions are drawn from it.

4. The finite-J audit in Section 7.14 is important and, in fact, shows that the remaining dimension contrast is still confounded. Calibrating the reference ensemble does not calibrate the nested subensembles whose errors are used in fitting the slope, and the manuscript finds systematic dimension-dependent deviations of $\mathrm{ESS}_0(J)/J$ from the target. The subsequent experiment that chooses a different β_J at every J controls this quantity, but then the fitted curve is obtained from a family of different algorithms rather than from one algorithm at different ensemble sizes.

The paper appropriately acknowledges this difficulty, but it also means that the remaining change of 0.054, 0.059, and 0.078 cannot yet be interpreted cleanly as a dimension effect after controlling concentration. I suggest reorganizing the presentation so that this limitation is stated before, rather than after, the principal numerical claim. More importantly, the fixed-β experiment suggested above seems necessary to determine what part of the effect survives under a design that is simultaneously interpretable algorithmically and reasonably well calibrated.

5. The "operating level" conclusion in Section 7.5 also needs to be weakened or recomputed. The statement that $\tau_0 \geq 0.1$ is sufficient in all 54 tested cells is based on the probe-calibrated

experiments. Section 7.13 subsequently demonstrates that the achieved reference concentration can differ substantially from the nominal target under precisely this calibration procedure. Thus $\tau_0 = 0.1$ is presently an operating level for the stated probe-calibration protocol, not necessarily an operating level in terms of the actual initial concentration. Either this experiment should be repeated using a defensible fixed-β calibration, or the interpretation should be restricted accordingly.

6. The uncertainty analysis for the headline dimension contrasts should be reconsidered. Table 3 bootstraps the twelve objective–initial law combinations and treats these combinations as the resampling units. These are selected design conditions rather than a random sample from a defined population of objective functions and initial laws. It is therefore unclear what population-level confidence statement this bootstrap is intended to support.

   For the principal numerical comparisons, uncertainty arising from the independent simulation realizations seems more directly interpretable. Variation across the selected objectives and initial laws can then be reported separately as heterogeneity over the chosen test set. If the author intends to make an inference over a population of optimization problems, that population and the sampling mechanism would first need to be defined. At minimum, the main contrasts should be accompanied by the realization-level analysis rather than relying primarily on a bootstrap over twelve hand-selected design settings.

7. Some statements comparing the original probe-calibrated experiment with the later exact-calibration experiment are too strong. The manuscript explicitly notes that the two experiments were run with different random streams and are not seed-paired. Nevertheless, statements such as changing the calibration "moved the statistic by 0.300" and descriptions of this as the "measured effect" of the protocol invite a causal interpretation. The observed difference contains both the protocol change and the sampling variation between two independent computational experiments. Unless a paired rerun or an uncertainty analysis for the difference is provided, the wording should be changed accordingly.

8. There is a sign inconsistency in the definition of the fitted exponent. Section 7 writes

$$E_J \approx CJ^{-\gamma}$$

   while $\widehat{\gamma}$ is subsequently reported as approximately -1 and is described as the fitted slope of $\log E_J$ against $\log J$. If $\widehat{\gamma}$ denotes the slope, the natural expression is

$$E_J \approx CJ^{\gamma}, \qquad \gamma \approx -1.$$

   Alternatively, if the model is written as $CJ^{-\gamma}$, then the theoretical exponent is $\gamma = 1$. The notation should be made consistent throughout, particularly because several later quantities are defined using signs of $\widehat{\gamma}$.

9. The interpretation of the bootstrap interval in the rate-consistency criterion should also be clarified. The manuscript uses the upper endpoint of a "bootstrap 95% interval" and describes this as a one-sided reading of a one-sided theoretical bound. If the reported percentile interval is the usual two-sided 95% interval, its upper endpoint corresponds to the 97.5th percentile rather than a 95% one-sided limit. This may simply make the criterion more conservative, but the actual confidence level and construction should be stated consistently.

10. The manuscript would benefit from substantial shortening and reorganization. The history of withdrawn hypotheses, failed criteria, lost files, superseded analyses, and successive diagnostic experiments is valuable for reproducibility, but it currently occupies a large fraction of the main narrative. Much of this material could be moved to the supplementary document. The main paper would be considerably stronger if it concentrated on a small number of cleanly posed questions and experiments: the concentration dependence, the finite-J observability study under a fixed algorithm, and the comparison of the two diffusion regimes.

    The author's willingness to document corrections and negative results is commendable, but transparency does not require every stage of the exploratory process to remain in the main article. A shorter manuscript centered on experiments whose estimands and theoretical assumptions are fully aligned would be considerably easier to evaluate and would make the actual contribution much clearer.

In summary, I find the underlying question interesting and I believe there may be a publishable contribution here. However, the incorrect classification of Ackley affects one of the manuscript's central conceptual claims, while the realization-dependent calibration of β makes the main corrected experiment difficult to interpret relative to the fixed-parameter theory. Since resolving these issues appears likely to require new computations rather than only editorial revision, I recommend rejection of the present manuscript. I would, however, encourage the author to submit a substantially revised new version if the theoretical assumptions are corrected and the central experiments are repeated under a fixed and independently calibrated parameter design.